CLIENT CODE : C000029969

CLIENT'S NAME AND ADDRESS :
AGARWAL PATHOLOGY
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PATIENT NAME : RADHE SHYAM MITTAL PATIENT ID : **RADHM231160102**

ACCESSION NO : **0102TK001417** AGE : 60 Years SEX : Male DATE OF BIRTH :

DRAWN : RECEIVED : 23/11/2020 09:54 REPORTED : 23/11/2020 13:31

REFERRING DOCTOR : DR. DYANESH GAWARE CLIENT PATIENT ID :

Test Report Status **Final**	Results		Biological Reference Interval	Units
HAEMATOLOGY				
COMPLETE BLOOD COUNT, EDTA WHOLE BLOOD/SMEAR				
BLOOD COUNTS				
HEMOGLOBIN	**11.7**	**Low**	13.0 - 17.0	g/dL
RED BLOOD CELL COUNT	**4.22**	**Low**	4.5 - 5.5	mil/µL
WHITE BLOOD CELL COUNT	6.70		4.0 - 10.0	thou/µL
PLATELET COUNT	234		150 - 410	thou/µL
RBC AND PLATELET INDICES				
HEMATOCRIT	**37.8**	**Low**	40.0 - 50.0	%
MEAN CORPUSCULAR VOL	89.6		83 - 101	fL
MEAN CORPUSCULAR HGB.	27.7		27.0 - 32.0	pg
MEAN CORPUSCULAR HEMOGLOBIN CONCENTRATION	**31.0**	**Low**	31.5 - 34.5	g/dL
RED CELL DISTRIBUTION WIDTH	13.7		11.6 - 14.0	%
MEAN PLATELET VOLUME	**11.6**	**High**	6.8 - 10.9	fL
WBC DIFFERENTIAL COUNT				
SEGMENTED NEUTROPHILS	56		40 - 80	%
EOSINOPHILS	04		1 - 6	%
LYMPHOCYTES	34		20 - 40	%
MONOCYTES	06		2 - 10	%
BASOPHILS	00		< 1 - 2	%
BAND (STAB) CELLS	00		0 - 5	%
PERIPHERAL SMEAR EXAM, EDTA WHOLE BLOOD				
RBC	Microcytic Hypochromic with anisocytosis . No heamparasite seen.			
WBC	Normal in morphology & distribution.No immature cells seen.			
PLATELETS	Adequate in number & normal in morphology.			
IMPRESSION	MICROCYTIC HYPOCHROMIC BLOOD PICTURE			

Interpretation(s)
BLOOD COUNTS-
The cell morphology is well preserved for 24hrs. However after 24-48 hrs a progressive increase in MCV and HCT is observed leading to a decrease in MCHC. A direct smear is recommended for an accurate differential count and for examination of RBC morphology.
RBC AND PLATELET INDICES-
The cell morphology is well preserved for 24hrs. However after 24-48 hrs a progressive increase in MCV and HCT is observed leading to a decrease in MCHC. A direct smear is recommended for an accurate differential count and for examination of RBC morphology.

BIO CHEMISTRY

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ACCESSION NO : **0102TK001417** AGE : 60 Years SEX : Male DATE OF BIRTH :

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REFERRING DOCTOR : DR. DYANESH GAWARE CLIENT PATIENT ID :

Test Report Status **Final**	**Results**		**Biological Reference Interval**	**Units**
SERUM BLOOD UREA NITROGEN				
BLOOD UREA NITROGEN	19		6 - 20	mg/dL
METHOD : SPECTROPHOTOMETRY, UREASE UV				
CREATININE, SERUM				
CREATININE	**1.77**	**High**	0.90 - 1.30	mg/dL
METHOD : SPECTROPHOTOMETRIC, JAFFE'S KINETICS				
ELECTROLYTES (NA/K/CL), SERUM				
SODIUM	144		136 - 145	mmol/L
METHOD : IMT - INDIRECT				
POTASSIUM	**5.30**	**High**	3.50 - 5.10	mmol/L
METHOD : IMT - INDIRECT				
CHLORIDE	**108**	**High**	98 - 107	mmol/L
METHOD : IMT - INDIRECT				
LIPID PROFILE PLUS				
CORONARY RISK PROFILE (LIPID PROFILE), SERUM.				
CHOLESTEROL	123		< 200 Desirable 200 - 239 Borderline High >/= 240 High	mg/dL
METHOD : ENZYMATIC/COLORIMETRIC,CHOLESTEROL OXIDASE, ESTERASE, PEROXIDASE				
TRIGLYCERIDES	**205**	**High**	< 150 Normal 150 - 199 Borderline High 200 - 499 High >/=500 Very High	mg/dL
METHOD : GPO - POD METHOD				
HDL CHOLESTEROL	**10**	**Low**	< 40 Low >/=60 High	mg/dL
METHOD : DIRECT ENZYME CLEARANCE				
DIRECT LDL CHOLESTEROL	93		< 100 Optimal 100 - 129 Near or above optimal 130 - 159 Borderline High 160 - 189 High >/= 190 Very High	mg/dL
METHOD : DIRECT ENZYME CLEARANCE				
NON HDL CHOLESTEROL	113		Desirable: Less than 130 Above Desirable: 130 - 159 Borderline High: 160 - 189 High: 190 - 219 Very high: > or = 220	mg/dL

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CHOL/HDL RATIO	**12.3**	**High**	3.3 - 4.4 Low Risk 4.5 - 7.0 Average Risk 7.1 - 11.0 Moderate Risk > 11.0 High Risk	
LDL/HDL RATIO	**9.3**	**High**	0.5 - 3.0 Desirable/Low Risk 3.1 - 6.0 Borderline/Moderate Risk >6.0 High Risk	
METHOD : CALCULATED PARAMETER				
VERY LOW DENSITY LIPOPROTEIN	**41.0**	**High**	</= 30.0	mg/dL
METHOD : CALCULATED PARAMETER				
GLUCOSE, FASTING, PLASMA				
GLUCOSE, FASTING, PLASMA	**101**	**High**	74 - 99	mg/dL
METHOD : SPECTROPHOTOMETRY HEXOKINASE				
GLYCOSYLATED HEMOGLOBIN, BLOOD				
GLYCOSYLATED HEMOGLOBIN (HBA1C)	**5.8**	**High**	Non-diabetic: < 5.7 Pre-diabetics: 5.7 - 6.4 Diabetics: > or = 6.5 ADA Target: 7.0 Action suggested: > 8.0	%
METHOD : AFFINITY COLUMN CHROMATOGRAPHY METHOD				
MEAN PLASMA GLUCOSE	**120**	**High**	< 116.0	mg/dL
LIVER FUNCTION TEST (WITHOUT GGT)				
ASPARTATE AMINOTRANSFERASE, SERUM				
ASPARTATE AMINOTRANSFERASE (AST/SGOT)	28		15 - 37	U/L
METHOD : SPECTROPHOTOMETRY, UV WITH PYRIDOXAL -5-PHOSPHATE				
ALANINE AMINOTRANSFERASE, SERUM				
ALANINE AMINOTRANSFERASE (ALT/SGPT)	21		< 45.0	U/L
METHOD : SPECTROPHOTOMETRY, UV WITH PYRIDOXAL -5-PHOSPHATE				
ALKALINE PHOSPHATASE, SERUM				
ALKALINE PHOSPHATASE	**23**	**Low**	30 - 120	U/L
METHOD : SPECTROPHOTOMETRY, P-NPP (AMP BUFFER)				
LACTATE DEHYDROGENASE, SERUM				
LACTATE DEHYDROGENASE	131		100 - 190	U/L
METHOD : SPECTROPHOTOMETRY, MODIFIED ENZYMATIC LACTATE - PYRUVATE				
BILIRUBIN (TOTAL, DIRECT, INDIRECT), SERUM				
BILIRUBIN, TOTAL	0.40		0.2 - 1.0	mg/dL
METHOD : SPECTROPHOTOMETRY, MODIFIED DIAZO METHOD (JENDRASSIK AND GROF)				
BILIRUBIN, DIRECT	0.20		0.0 - 0.2	mg/dL

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Test Report Status **Final**	**Results**	**Biological Reference Interval**	**Units**
METHOD : SPECTROPHOTOMETRY, MODIFIED DIAZO METHOD (JENDRASSIK AND GROF)			
BILIRUBIN, INDIRECT	0.20	0.1 - 1.0	mg/dL
METHOD : CALCULATED PARAMETER			
TOTAL PROTEIN,ALBUMIN,GLOBULIN, SERUM			
TOTAL PROTEIN	7.3	6.4 - 8.2	g/dL
METHOD : SPECTROPHOTOMETRY, MODIFIED BIURET			
ALBUMIN	3.7	3.4 - 5.0	g/dL
METHOD : SPECTROPHOTOMETRY, MODIFIED BIURET			
GLOBULIN	3.6	2.0 - 4.1	g/dL
METHOD : CALCULATED PARAMETER			
ALBUMIN/GLOBULIN RATIO	1.0	1.0 - 2.1	Ratio
METHOD : CALCULATED PARAMETER			
URIC ACID, SERUM			
URIC ACID	6.6	3.5 - 7.2	mg/dL
METHOD : SPECTROPHOTOMETRY, URICASE			
CALCIUM, SERUM			
CALCIUM	9.2	8.5 - 10.1	mg/dL
METHOD : O-CRESOLPHTHALEIN COMPLEXONE			

Interpretation(s)
SERUM BLOOD UREA NITROGEN-
Causes of Increased levels
Pre renal
• High protein diet, Increased protein catabolism, GI haemorrhage, Cortisol, Dehydration, CHF Renal
• Renal Failure
Post Renal
• Malignancy, Nephrolithiasis, Prostatism

Causes of decreased levels
• Liver disease
• SIADH.
CREATININE, SERUM-
Higher than normal level may be due to:
• Blockage in the urinary tract
• Kidney problems, such as kidney damage or failure, infection, or reduced blood flow
• Loss of body fluid (dehydration)
• Muscle problems, such as breakdown of muscle fibers
• Problems during pregnancy, such as seizures (eclampsia)), or high blood pressure caused by pregnancy (preeclampsia)

Lower than normal level may be due to:
• Myasthenia Gravis
• Muscular dystrophy
ELECTROLYTES (NA/K/CL), SERUM-
Sodium levels are Increased in dehydration, cushing's syndrome, aldosteronism & decreased in Addison's disease, hypopituitarism,liver disease. Hypokalemia (low K) is common in vomiting, diarrhea, alcoholism, folic acid deficiency and primary aldosteronism. Hyperkalemia may be seen in end-stage renal failure, hemolysis, trauma, Addison's disease, metabolic acidosis, acute starvation, dehydration, and with rapid K infusion.Chloride is increased in dehydration, renal tubular acidosis (hyperchloremia metabolic acidosis), acute renal failure, metabolic acidosis associated with prolonged diarrhea and loss of sodium bicarbonate, diabetes insipidus, adrenocortical hyperfuction, salicylate intoxication and with excessive infusion of isotonic saline or extremely high dietary intake of salt.Chloride is decreased in overhydration, chronic respiratory acidosis, salt-losing nephritis, metabolic alkalosis, congestive heart failure, Addisonian crisis, certain types of metabolic acidosis, persistent gastric secretion and prolonged vomiting,

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CORONARY RISK PROFILE (LIPID PROFILE), SERUM.-
Serum cholesterol is a blood test that can provide valuable information for the risk of coronary artery disease This test can help determine your risk of the build up of plaques in your arteries that can lead to narrowed or blocked arteries throughout your body (atherosclerosis). High cholesterol levels usually don't cause any signs or symptoms, so a cholesterol test is an important tool. High cholesterol levels often are a significant risk factor for heart disease and important for diagnosis of hyperlipoproteinemia, atherosclerosis, hepatic and thyroid diseases.

Serum Triglyceride are a type of fat in the blood. When you eat, your body converts any calories it doesn't need into triglycerides, which are stored in fat cells. High triglyceride levels are associated with several factors, including being overweight, eating too many sweets or drinking too much alcohol, smoking, being sedentary, or having diabetes with elevated blood sugar levels. Analysis has proven useful in the diagnosis and treatment of patients with diabetes mellitus, nephrosis, liver obstruction, other diseases involving lipid metabolism, and various endocrine disorders. In conjunction with high density lipoprotein and total serum cholesterol, a triglyceride determination provides valuable information for the assessment of coronary heart disease risk.It is done in fasting state.

High-density lipoprotein (HDL) cholesterol. This is sometimes called the ""good"" cholesterol because it helps carry away LDL cholesterol, thus keeping arteries open and blood flowing more freely.HDL cholesterol is inversely related to the risk for cardiovascular disease. It increases following regular exercise, moderate alcohol consumption and with oral estrogen therapy. Decreased levels are associated with obesity, stress, cigarette smoking and diabetes mellitus.

SERUM LDL The small dense LDL test can be used to determine cardiovascular risk in individuals with metabolic syndrome or established/progressing coronary artery disease, individuals with triglyceride levels between 70 and 140 mg/dL, as well as individuals with a diet high in trans-fat or carbohydrates. Elevated sdLDL levels are associated with metabolic syndrome and an 'atherogenic lipoprotein profile', and are a strong, independent predictor of cardiovascular disease.
Elevated levels of LDL arise from multiple sources. A major factor is sedentary lifestyle with a diet high in saturated fat. Insulin-resistance and pre-diabetes have also been implicated, as has genetic predisposition. Measurement of sdLDL allows the clinician to get a more comprehensive picture of lipid risk factors and tailor treatment accordingly. Reducing LDL levels will reduce the risk of CVD and MI.

Non HDL Cholesterol - Adult treatment panel ATP III suggested the addition of Non-HDL Cholesterol as an indicator of all atherogenic lipoproteins (mainly LDL and VLDL). NICE guidelines recommend Non-HDL Cholesterol measurement before initiating lipid lowering therapy. It has also been shown to be a better marker of risk in both primary and secondary prevention studies.

Recommendations:
Results of Lipids should always be interpreted in conjunction with the patient's medical history, clinical presentation and other findings.

NON FASTING LIPID PROFILE includes Total Cholesterol, HDL Cholesterol and calculated non-HDL Cholesterol. It does not include triglycerides and may be best used in patients for whom fasting is difficult.
GLUCOSE, FASTING, PLASMA-
ADA 2012 guidelines for adults as follows:
Pre-diabetics: 100 - 125 mg/dL
Diabetic: > or = 126 mg/dL

(Ref: Tietz 4th Edition & ADA 2012 Guidelines)
GLYCOSYLATED HEMOGLOBIN, BLOOD-
Glycation is nonenzymatic addition of sugar residue to amino groups of proteins. HbA1C is formed by the condensation of glucose with n-terminal valine residue of each beta chain of hb a to form an unstable schif base. It is the major fraction, constituting approximately 80% of HbA1.
Formation of glycated hemoglobin (GHb) is essentially irreversible and the concentration in the blood depends on both the lifespan of the red blood cells (RBC) (120 days) and the blood glucose concentration. The GHB concentration represents the integrated values for glucose aver the period of 6 to 8 weeks. GHb values are free of day to day glucose fluctuations and are unaffected by recent exercise or food ingestion. Concentration of plasma glucose concentration in GHb depends on the time interval, with more recent values providing a larger contribution than earlier values.

The interpretation of GHb depends on RBC having a normal life span. Patients with hemolytic disease or other conditions with shortened RBC survival exhibit a substantial reduction of GHb. High GHb have been reported in iron deficiency anemia .

GHb has been firmly established as an index of long term blood glucose concentrations and as a measure of the risk for the development of complications in patients with diabetes mellitus. The absolute risk of retinopathy and nephropathy are directly proportional to the mean of HbA1C.

"Targets should be individualized; More or less stringent glycemic goals may be appropriate for individual patients. Goals should be individualized based on duration of diabetes, age/life expectancy, comorbid conditions, known CVD or advanced microvascular complications, hypoglycemia unawareness, and individual patient considerations."
ASPARTATE AMINOTRANSFERASE, SERUM-
Aminotransferase (AST) is an enzyme found in various parts of the body .AST is found in the liver, heart, skeletal muscle, kidneys, brain, and red blood cells, and it is commonly measured clinically as a marker for liver health. AST levels increase during chronic viral hepatitis, blockage of the bile duct, cirrhosis of the liver, liver cancer, kidney failure, hemolytic anemia, pancreatitis, hemochromatosis. AST levels may also increase after a heart attack or strenuous activity.
ALANINE AMINOTRANSFERASE, SERUM-
Alanine aminotransferase (ALT) test measures the amount of this enzyme in the blood. ALT is found mainly in the liver, but also in smaller amounts in the kidneys, heart, muscles, and pancreas. It is commonly measured as a part of a diagnostic evaluation of hepatocellular injury, to determine liver health. . AST levels increase during acute hepatitis, sometimes due to a viral infection, ischemia to the liver, chronic hepatitis, obstruction of bile ducts, cirrhosis.
ALKALINE PHOSPHATASE, SERUM-
Alkaline phosphatase (ALP) is a protein found in almost all body tissues. Tissues with higher amounts of ALP include the liver, bile ducts, and bone. Elevated Alkaline Phosphaqtase levels are seen in Biliary obstruction,Osteoblastic bone tumors, osteomalacia, hepatitis, Hyperparathyroidism,Leukemia, Lymphoma,Paget's disease,Rickets,Sarcoidosis etc. Lower-than-normal ALP levels seen in Hypophosphatasia, Malnutrition, Protein deficiency,Wilson's disease .
LACTATE DEHYDROGENASE, SERUM-
LDH is an enzyme that helps in energy production. It is present in almost all of the tissues in the body and its levels rise in response to cell damage. LDH levels help to

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diagnose lung disease, lymphoma, anemia, and liver disease. They also help determine how well chemotherapy is working .A higher-than-normal level may indicate:Blood flow deficiency (ischemia), Heart attack, Hemolytic anemia, Infectious mononucleosis, Liver disease (for example, hepatitis),Low blood pressure,Muscle injury, muscular dystrophy, New abnormal tissue formation usually cancer, Pancreatitis and Stroke.
BILIRUBIN (TOTAL, DIRECT, INDIRECT), SERUM-
Bilirubin is a yellowish pigment found in bile and is a breakdown product of normal heme catabolism. Bilirubin is excreted in bile and urine, and elevated levels may give yellow discoloration in jaundice.Elevated levels results from increased bilirubin production (eg, hemolysis and ineffective erythropoiesis), decreased bilirubin excretion (eg, obstruction and hepatitis), and abnormal bilirubin metabolism (eg, hereditary and neonatal jaundice). Conjugated (direct) bilirubin is elevated more than unconjugated (indirect) bilirubin in Viral hepatitis, Drug reactions, Alcoholic liver disease Conjugated (direct) bilirubin is also elevated more than unconjugated (indirect) bilirubin when there is some kind of blockage of the bile ducts like in Gallstones getting into the bile ducts, tumors & Scarring of the bile ducts. Increased unconjugated (indirect) bilirubin may be a result of Hemolytic or pernicious anemia, Transfusion reaction & a common metabolic condition termed Gilbert syndrome, due to low levels of the enzyme that attaches sugar molecules to bilirubin.

Total Bili-
Source: Wallach"s Interpretation of Diagnostic tests, 9th ed

Direct Bili -
Source: Tietz Text book of Clinical Chemistry & Molecular Diagnostics, 4th ed
TOTAL PROTEIN,ALBUMIN,GLOBULIN, SERUM-
Serum total protein,also known as total protein, is a biochemical test for measuring the total amount of protein in serum..Protein in the plasma is made up of albumin and globulin.

Higher-than-normal levels may be due to: Chronic inflammation or infection, including HIV and hepatitis B or C, Multiple myeloma, Waldenstrom's disease

Lower-than-normal levels may be due to: Agammaglobulinemia, Bleeding (hemorrhage), Burns, Glomerulonephritis, Liver disease, Malabsorption, Malnutrition, Nephrotic syndrome, Protein-losing enteropathy etc.Human serum albumin is the most abundant protein in human blood plasma. It is produced in the liver. Albumin constitutes about half of the blood serum protein. Low blood albumin levels (hypoalbuminemia) can be caused by:Liver disease like cirrhosis of the liver, nephrotic syndrome, protein-losing enteropathy, Burns, hemodilution, increased vascular permeability or decreased lymphatic clearance,malnutrition and wasting etc.
URIC ACID, SERUM-
Causes of Increased levels
Dietary
• High Protein Intake.
• Prolonged Fasting,
• Rapid weight loss.
Gout
Lesch nyhan syndrome.
Type 2 DM.
Metabolic syndrome.

Causes of decreased levels
• Low Zinc Intake
• OCP's
• Multiple Sclerosis

Nutritional tips to manage increased Uric acid levels
• Drink plenty of fluids
• Limit animal proteins
• High Fibre foods
• Vit C Intake
• Antioxidant rich foods
CALCIUM, SERUM-
Commom causes of decreased value of calcium (hypocalcemia) are chronic renal failure, hypomagnesemia and hypoalbuminemia.

Hypercalcemia (increased value of calcium) can be caused by increased intestinal absorbtion (vitamin d intoxication), increased skeletal reasorption (immobilization), or a combination of mechanisms (primary hyperparathyroidism). Primary hyperparathyroidism and malignancy accounts for 90-95% of all cases of hypercalcemia.

Values of total calcium is affected by serum proteins, particularly albumin thus, latter's value should be taken into account when interpreting serum calcium levels. The following regression equation may be helpful.
Corrected total calcium (mg/dl)= total calcium (mg/dl) + 0.8 (4- albumin [g/dl])*
because regression equations vary among group of patients in different physiological and pathological conditions, mathematical corrections are only approximations. The possible mathematical corrections should be replaced by direct determination of free calcium by ISE (available with srl) a common and important source of preanalytical error in the measurement of calcium is prolonged torniquet application during sampling. Thus, this along with fist clenching should be avoided before phlebotomy.

ENDOCRINOLOGY

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THYROID PANEL BY CHEMILUMINESCENCE, SERUM

T3	61.54	60.0 - 181.0	ng/dL
T4	9.00	4.5 - 10.9	µg/dl
TSH 3RD GENERATION	3.536	0.55 - 4.78	µIU/mL

Interpretation(s)
THYROID PANEL BY CHEMILUMINESCENCE, SERUM-
Triiodothyronine T3 , is a thyroid hormone. It affects almost every physiological process in the body, including growth, development, metabolism, body temperature, and heart rate. Production of T3 and its prohormone thyroxine (T4) is activated by thyroid-stimulating hormone (TSH), which is released from the pituitary gland. Elevated concentrations of T3, and T4 in the blood inhibit the production of TSH.
Thyroxine T4, Thyroxine's principal function is to stimulate the metabolism of all cells and tissues in the body. Excessive secretion of thyroxine in the body is hyperthyroidism, and deficient secretion is called hypothyroidism. Most of the thyroid hormone in blood is bound to transport proteins. Only a very small fraction of the circulating hormone is free and biologically active.
In primary hypothyroidism, TSH levels are significantly elevated, while in secondary and tertiary hypothyroidism, TSH levels are low.
Below mentioned are the guidelines for Pregnancy related reference ranges for Total T4, TSH & Total T3

Levels in Pregnancy	TOTAL T4 (µg/dL)	TSH3G (µIU/mL)	TOTAL T3 (ng/dL)
First Trimester	6.6 - 12.4	0.1 - 2.5	81 - 190
2nd Trimester	6.6 - 15.5	0.2 - 3.0	100 - 260
3rd Trimester	6.6 - 15.5	0.3 - 3.0	100 - 260

Below mentioned are the guidelines for age related reference ranges for T3 and T4.

T3 (ng/dL)	T4 (µg/dL)
New Born: 75 - 260	1-3 day: 8.2 - 19.9
.	1 Week: 6.0 - 15.9

NOTE: TSH concentrations in apparently normal euthyroid subjects are known to be highly skewed, with a strong tailed distribution towards higher TSH values. This is well documented in the pediatric population including the infant age group.
Kindly note: Method specific reference ranges are appearing on the report under biological reference range.

Reference:
1. Burtis C.A., Ashwood E. R. Bruns D.E. Teitz textbook of Clinical Chemistry and Molecular Diagnostics, 4th Edition.
2. Gowenlock A.H. Varley's Practical Clinical Biochemistry, 6th Edition.
3. Behrman R.E. Kilegman R.M., Jenson H. B. Nelson Text Book of Pediatrics, 17th Edition

CLINICAL PATH

URINALYSIS

COLOR	PALE YELLOW	
APPEARANCE	CLEAR	
PH	6.0	4.7 - 7.5
SPECIFIC GRAVITY	1.025	1.003 - 1.035
GLUCOSE	NOT DETECTED	NOT DETECTED
PROTEIN	NOT DETECTED	NOT DETECTED
KETONES	NOT DETECTED	NOT DETECTED
BLOOD	NOT DETECTED	NOT DETECTED
BILIRUBIN	NOT DETECTED	NOT DETECTED
UROBILINOGEN	NORMAL	NORMAL

CLIENT CODE : C000029969

CLIENT'S NAME AND ADDRESS :
AGARWAL PATHOLOGY
49, KAMLA NEHRU MARG, AGARWAL DIAGNOSTICS, FIRST FLOOR,
FREEGANJ,

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CIN - U74899PB1995PLC045956
Email : srlujjain@gmail.com

PATIENT NAME : RADHE SHYAM MITTAL PATIENT ID : **RADHM231160102**

ACCESSION NO : **0102TK001417** AGE : 60 Years SEX : Male DATE OF BIRTH :

DRAWN : RECEIVED : 23/11/2020 09:54 REPORTED : 23/11/2020 13:31

REFERRING DOCTOR : DR. DYANESH GAWARE CLIENT PATIENT ID :

Test Report Status **Final**	**Results**	**Biological Reference Interval**	**Units**
NITRITE	NOT DETECTED	NOT DETECTED	
WBC	1-2	0-5	/HPF
EPITHELIAL CELLS	1-2	0-5	/HPF
RED BLOOD CELLS	NOT DETECTED	NOT DETECTED	/HPF
CASTS	NOT DETECTED		
CRYSTALS	NOT DETECTED		
BACTERIA	DETECTED (OCCASIONAL)	NOT DETECTED	

Interpretation(s)
URINALYSIS-Routine urine analysis assists in screening and diagnosis of various metabolic, urological, kidney and liver disorders
Protein: Elevated proteins can be an early sign of kidney disease. Urinary protein excretion can also be temporarily elevated by strenuous exercise, orthostatic proteinuria, dehydration, urinary tract infections and acute illness with fever
Glucose: Uncontrolled diabetes mellitus can lead to presence of glucose in urine. Other causes include pregnancy, hormonal disturbances, liver disease and certain medications.
Ketones: Uncontrolled diabetes mellitus can lead to presence of ketones in urine. Ketones can also be seen in starvation, frequent vomiting, pregnancy and strenuous exercise.
Blood: Occult blood can occur in urine as intact erythrocytes or haemoglobin, which can occur in various urological, nephrological and bleeding disorders.
Leukocytes: An increase in leukocytes is an indication of inflammation in urinary tract or kidneys. Most common cause is bacterial urinary tract infection.
Nitrite: Many bacteria give positive results when their number is high. Nitrite concentration during infection increases with length of time the urine specimen is retained in bladder prior to collection.
pH: The kidneys play an important role in maintaining acid base balance of the body. Conditions of the body producing acidosis/ alkalosis or ingestion of certain type of food can affect the pH of urine.
Specific gravity: Specific gravity gives an indication of how concentrated the urine is. Increased specific gravity is seen in conditions like dehydration, glycosuria and proteinuria while decreased specific gravity is seen in excessive fluid intake, renal failure and diabetes insipidus.
Bilirubin: In certain liver diseases such as biliary obstruction or hepatitis, bilirubin gets excreted in urine.
Urobilinogen: Positive results are seen in liver diseases like hepatitis and cirrhosis and in cases of hemolytic anemia

****End Of Report****

Please visit www.srlworld.com for related Test Information for this accession

Dr. Chitra A Sinha
Pathologist